Amazon Web Services: Elastic Load Balancing (ELB)

Topics

- Open System Interconnection (OSI) Model
- What is Elastic Load Balancing?
- Types of Load Balancers
- Health Checks
- Application Load Balancers
- Network Load Balancers
- Classic Load Balancers
- Sticky Sessions
- Deregistration Delay

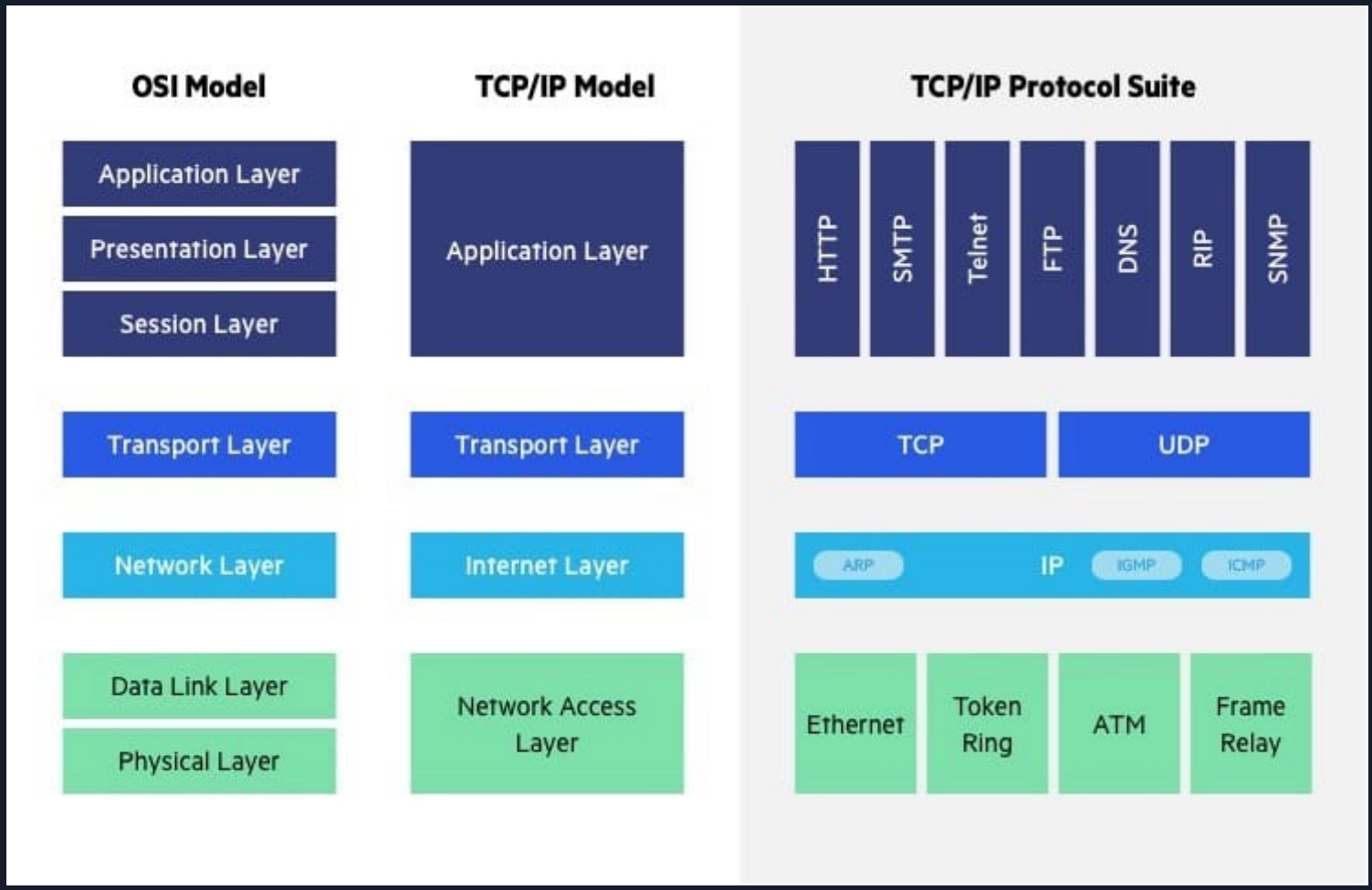
Open Systems Interconnection (OSI) Model

Open Systems Interconnection (OSI) Model

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

https://www.imperva.com/learn/application-security/osi-model/

OSI vs. TCP/IP Model

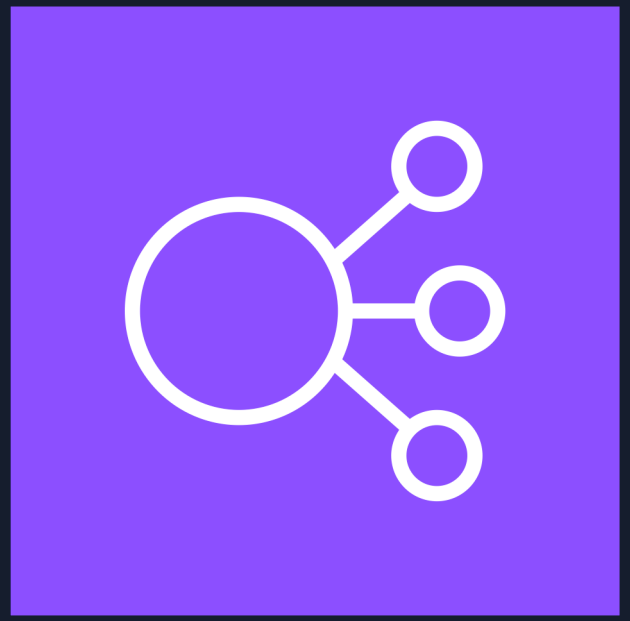


https://www.imperva.com/learn/application-security/osi-model/

Elastic Load Balancing

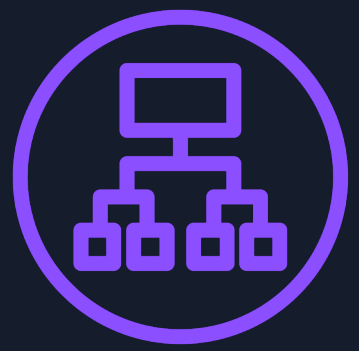
What is Elastic Load Balancing?

- Automatically distributes incoming traffic across multiple targets, such as Amazon EC2 instances
  - Can be done across multiple AZs

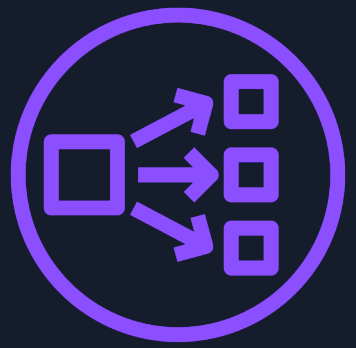


Elastic Load Balancing

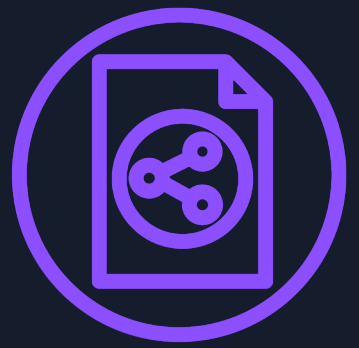
Types of Load Balancers



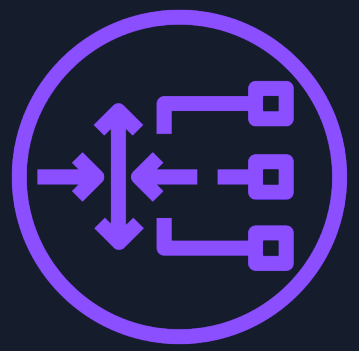
Application Load Balancer



Network Load Balancer



Classic Load Balancer

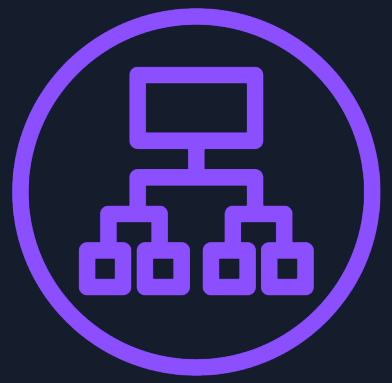


Gateway Load Balancer

https://aws.amazon.com/elasticloadbalancing/features/

Health Checks

- Health checks periodically send requests to load balancers' registered instances to test their status
- All AWS load balancers can be configured with health checks
- The status of healthy instances at the time of the health check is InService
- The status of unhealthy instances at the time of the health check is OutOfService



Application Load Balancer

Layer 7 Load Balancing

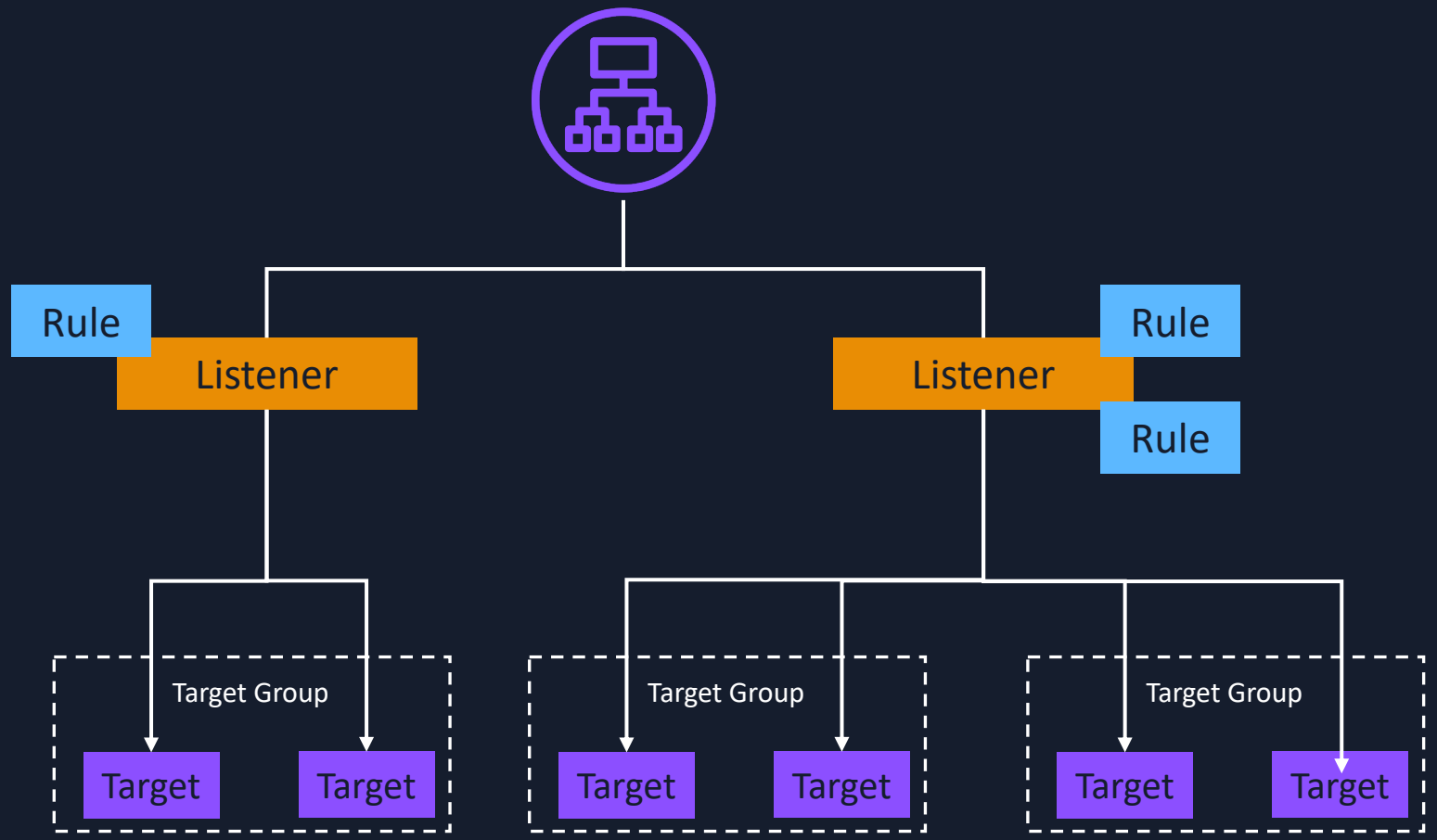
- An application load balancer functions at the Application Layer which is Layer 7 in the Open System Interconnection (OSI) model

- After the load balancer receives a request, it evaluates the listener rules in priority order to determine which rule to apply, and then selects a target from the target group for the rule action

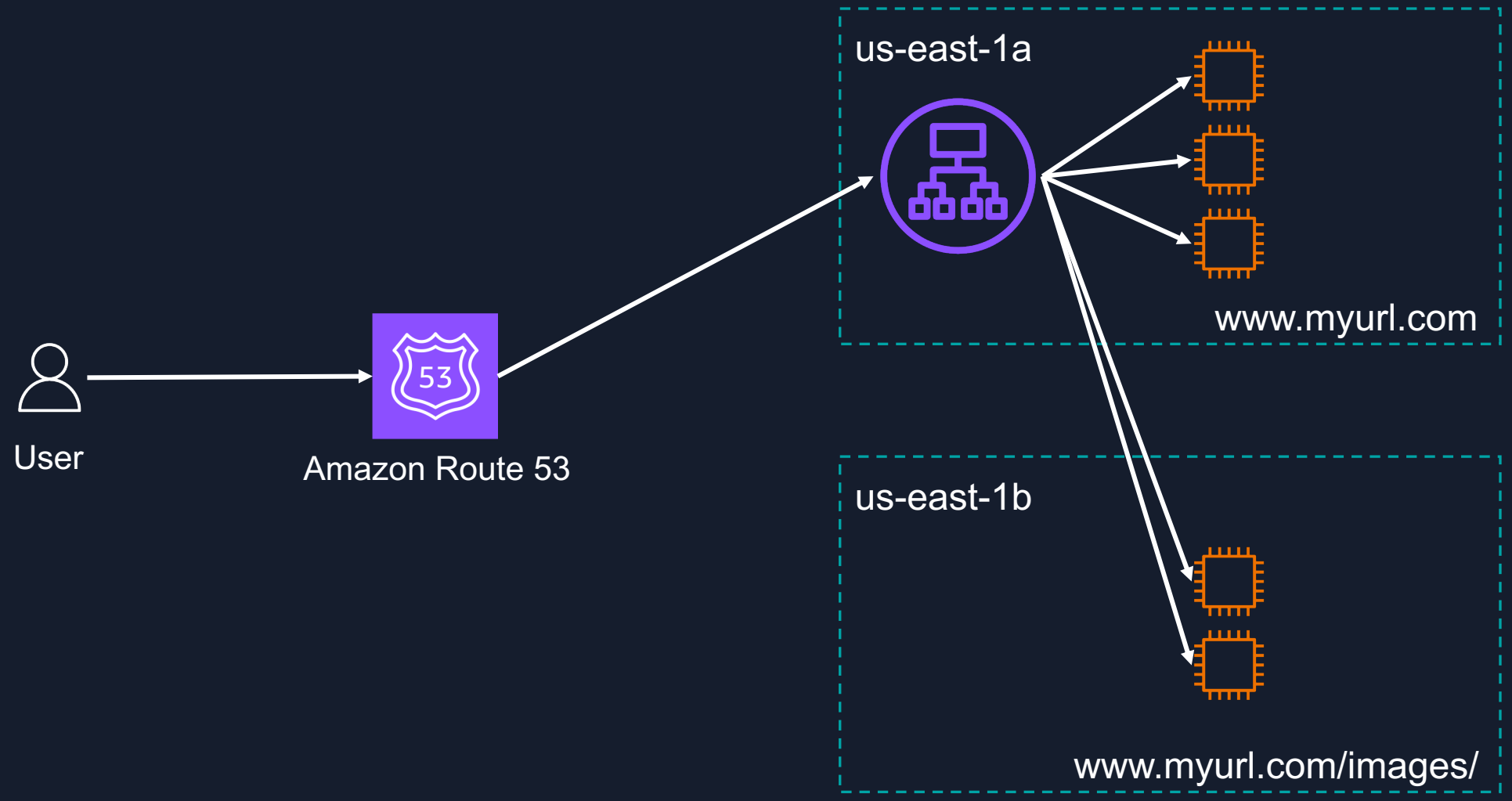
Listeners, Rules, and Target Groups

Listeners	Rules	Target Groups
• Check for connection requests from clients, using the protocol and port you configure • You define rules that determine how the load balancer routes requests to its registered targets • Each rules consists of a priority, one or more actions, and one or more conditions	• When the conditions for a rule are met, then its actions are performed. • You must define a default rule for each listener • You can optionally define additional rules	• Each target group routes requests to one or more registered targets, such as EC2 instances, using the protocol and port number you specify

Network Diagram



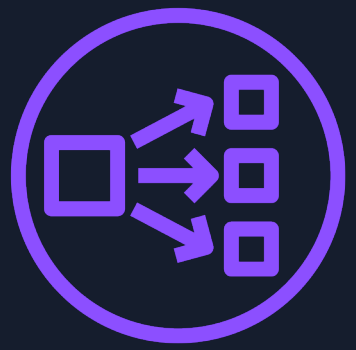
Path-based Routing



Limitations of Application Load Balancers

- Only supports HTTP and HTTPS
- To use an HTTPS listener, you must deploy at least one SSL/TLS server certificate on your load balancer
- The load balancer user a server certificate to terminate the frontend connection and then decrypt requests from clients before sending them to the targets

Demo: Application Load Balancer



Network Load Balancer

Layer 4 Load Balancing

- A Network Load Balancer functions at the fourth layer of the Open Systems Interconnection (OSI) model
- It can handle millions of requests per second
- Supported Ports : 1-65535
- Supported Protocols: TCP, TLS, UDP, TCP_UDP

Request Received, Listener, and Target Groups

Request Received	Listeners	Target Groups
• Receives request • Selects target from the target group • Attempts to open a TCP connection to the selected target on the port specified in the listener configuration	• Checks for connection requests from clients, using the protocol and configured port • Forwards the request to the target group • There are no rules, unlike with Application Load Balancers	• Each target group routes requests to one or more registered targets, such as EC2 instances, using the protocol and specified port number

Encryption

- You can use a TLS listener to offload the work of encryption and decryption to your load balancer so your applications can focus on their business logic

- If the listener protocol is TLS, you must deploy exactly one SSL server certificate on the listener

Classic Load Balancer

Classic Load Balancers

- Classic Load Balancers are the legacy load balancers
- Can load balance HTTP/ HTTPS applications and use Layer 7- specific features, such as X-Forwarded and sticky sessions
- Can also use strict Layer 4 load balancing for applications that rely purely on the TCP protocol

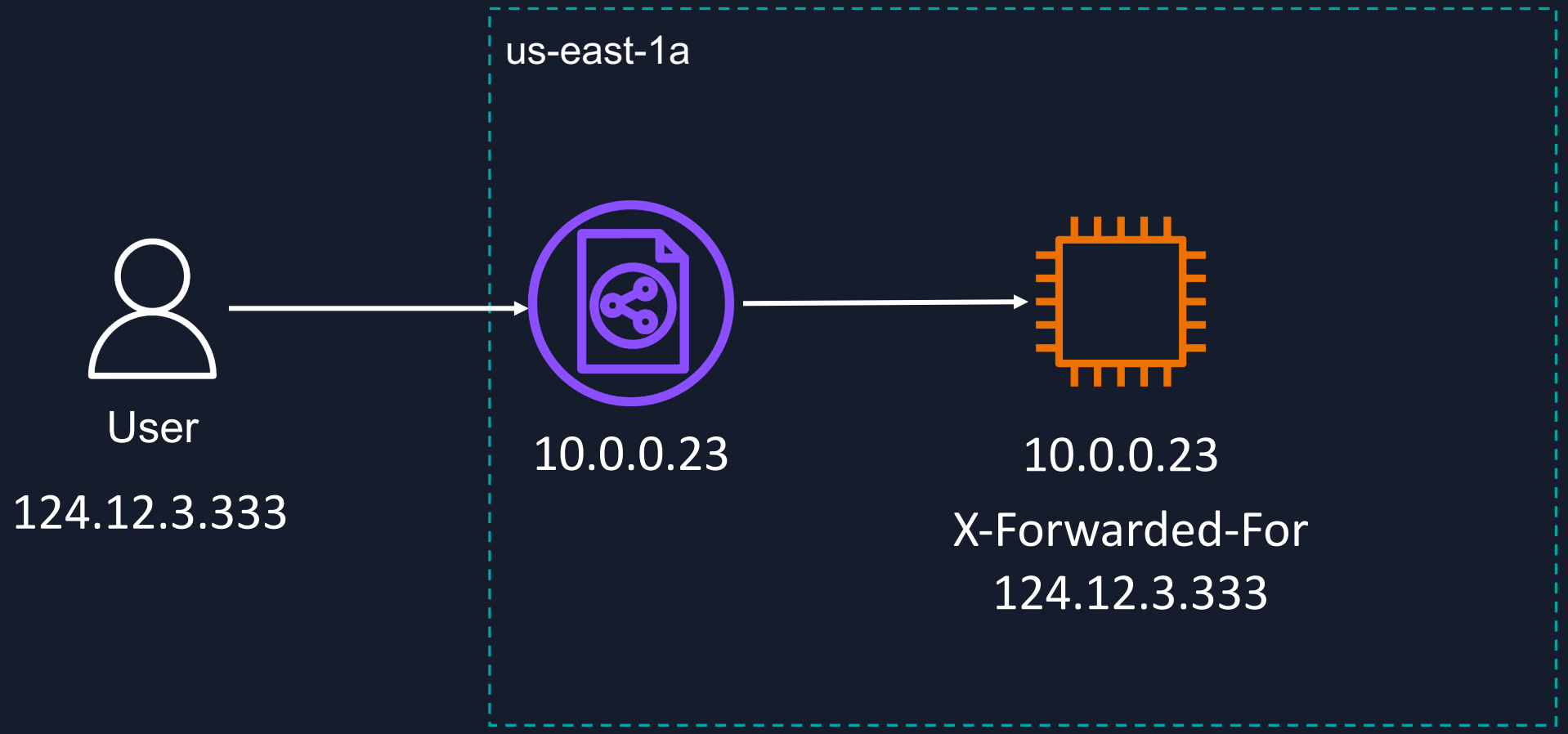
X-Forwarded-For

- When traffic is sent from a load balancer, the server access logs contain the IP address of the proxy or load balancer only

- To see the original IP address of the client, the X-Forwarded-For request header is used

```
Hypertext Transfer Protocol
> HEAD / HTTP/1.1\r\n
  Host: copyheader.com\r\n
  User-Agent: curl/7.51.0\r\n
  Accept: */*\r\n
  X-Forwarded-For: 10.0.11.209\r\n
  X-Forwarded-For-Port: 49956\r\n
  \r\n
  [Full request URI: http://copyheader.com/]
  [HTTP request 1/1]
  [Response in frame: 11]
```

Network Diagram



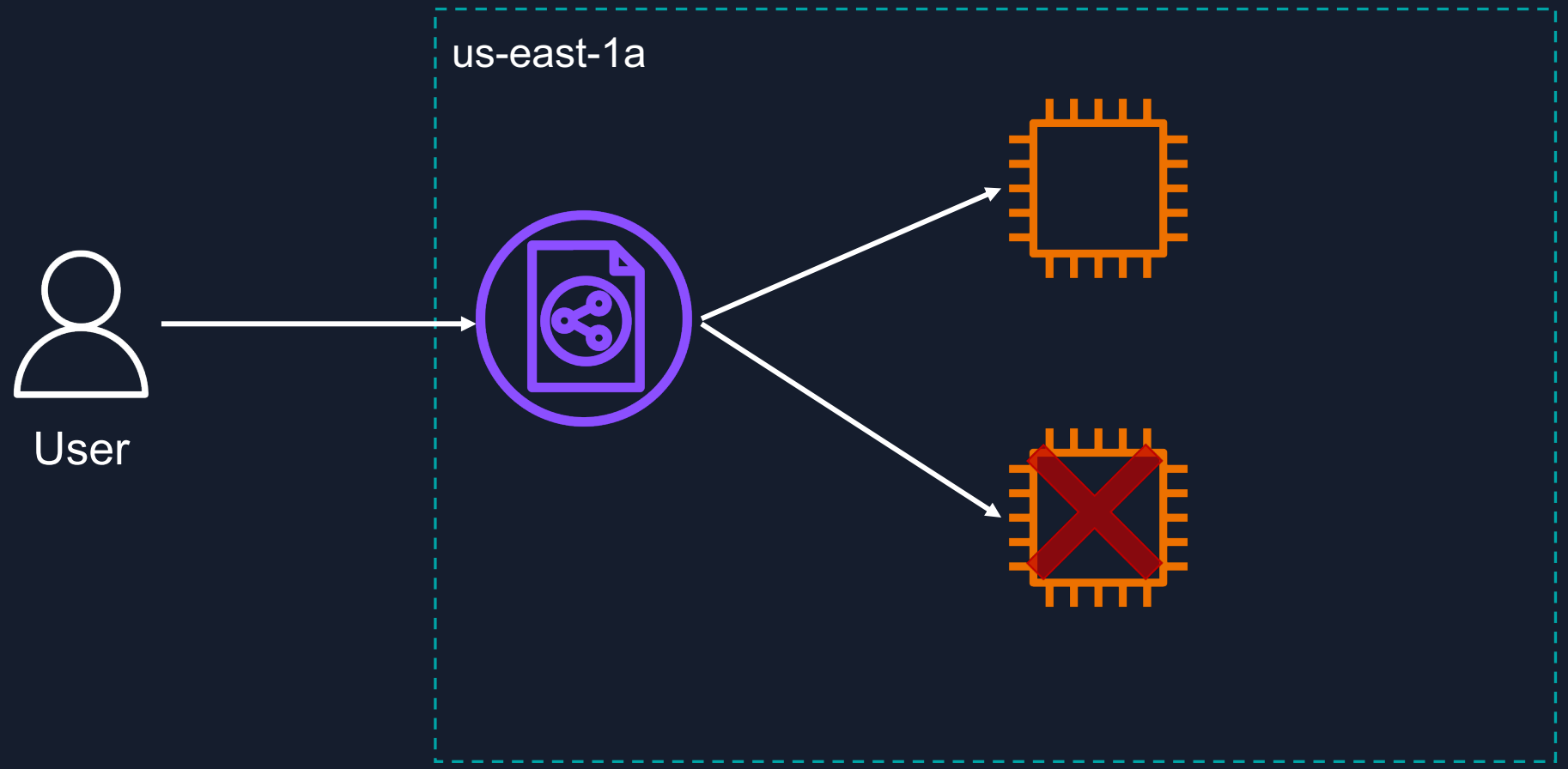
Demo: Classic Load Balancer

Sticky Sessions

What are sticky sessions?

- Classic Load Balancers route each request independently to the registered EC2 instance with the smallest load
- Sticky sessions allow you to bind a user’s session to a specific EC2 instance

Scaling Issues



Deregistration Delay

What is Deregistration Delay

- Allows load balancers to keep existing connections open if the EC2 instances are deregistered or become unhealthy
- Enables the load balancer to complete in-flight requests made to instances that are deregistering or unhealthy

Demo: Deregistration Delay

Checkpoint

Which of the following is NOT an AWS ELB load balancer?

1. Application Load Balancer (ALB)
2. Geolocation Load Balancer (GLB)
3. Network Load Balancer (NLB)
4. Classic Load Balancer (CLB)

Which kind of ELB load balancer would you use if you need extreme levels of performance?

1. Classic Load Balancer (CLB)
2. Performance Load Balancer (PLB)
3. Application Load Balancer (ALB)
4. Network Load Balancer (NLB)

What kind of ELB load balancer would you select if you need to route traffic based on the contents of the request?

1. Gateway Load Balancer (GLB)
2. Classic Load Balancer (CLB)
3. Application Load Balancer (ALB)
4. Network Load Balancer (NLB)

Which ELB setting would you enable to ensure each subsequent connection after the initial connection was routed to the same EC2 instance?

1. Biscuit sessions
2. Sticky sessions
3. Not needed, as this happens by default
4. This can't be done

Which layer of the OSI model does the NLB function on?

1. Layer 1
2. Layer 4
3. Layer 2
4. Layer 7

What benefit do you get from configuring health checks on an ELB load balancer?

1. Instances that fail the health check will not receive traffic.
2. When the ELB load balancer fails, AWS will automatically fix it.
3. This will prevent your instances from failing.
4. The health check will notify you when it's time for your next flu shot

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